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[priyaraman-ai-portfolio-76375761518.us-central1.run.app](#) (Portfolio)

Top Skills

AI Governance - Prompt Governance

AI Governance - Azure AI Content Safety / Guardrails

Enterprise LLM Platform Engineering

Certifications

Google 5 Day Gen AI Intensive

GenAI Bootcamp Certificate (Teal Squad)

OCP in DBA Track

LLM Course - LLM Fundamentals, Fine Tuning Language Models

Priya Raman

GenAI Architect & Engineer | Anthropic Claude on Azure AI Foundry & APIM • Multi-Agent Orchestration • Agentic RAG • Enterprise Reliability • LLM Fine-Tuning | Vertex AI & GCP | LLMOps | 18+ Years Tech Leadership

Aurora, Illinois, United States

Summary

A seasoned technology professional with an 18+ year track record of delivering enterprise-scale solutions across Healthcare, Retail, Supply Chain, and Banking. My career evolved from managing complex ERP systems and mission-critical mobile initiatives for CVS Health, McDonald's & Allstate into my current role architecting production-grade GenAI. I bridge deep systems engineering with advanced Agentic architectures, via a Post-Grad Program in AI/ML from UT Austin. I don't just build agents; I architect reliable reasoning systems. My hands-on experience spans the full GenAI lifecycle: Enterprise LLM Platform Engineering & Cost Optimization (Azure): Architecting production Anthropic Claude deployments on Azure AI Foundry for a banking client — prompt-cache optimization (71.8% peak cache-hit rate), two-pass multi-domain orchestration, and anti-hallucination evidence-sandboxing. Instrumented Application Insights telemetry and a pre-flight cost estimator for full session-cost visibility, plus metadata-driven model routing and APIM inter-service auth. Production LLM Deployment & Vertex AI: Deployed fine-tuned LLM to production using Vertex AI Endpoints with Gradio UI. Architected multi-agent systems and high-fidelity workflows to automate complex document ingestion and legacy-to-cloud migrations structured for Vertex AI Agent Engine. Built enterprise RAG using Vertex AI Search with dynamic filtering and Ranking API. LLM Fine-Tuning & MLOps : Fine-tuned Llama-3.1-8B achieving 331% performance gain via PEFT/LoRA. Full MLOps lifecycle: LLM-as-a-Judge evaluation, GGUF quantization, secure 3-tier GCP deployment using API gateway. Multi-Agent Orchestration & Agentic RAG : Distributed multi-agent systems using ADK 1.20, LangGraph, LlamaIndex with Mem0. 100% GAIA benchmark accuracy. Advanced RAG with RRF and hybrid search - BM25 + Semantic Search with FAISS for GoogleGenAI Embeddings Infrastructure & MCP: Leveraged

custom Model Context Protocol server to decouple heavy compute dependencies(TTS and MLX Inference)."Smart Tool" patterns reduced runner complexity by 80%. Generated 15 mins Podcasts in < 2 mins. Reliability & Observability: Engineered strict AI Guardrails - refusal heuristics for Digital Twin, Custom Manual Rubric Evaluators achieving 91% pass rates. "Telemetry Gravity" pattern for unified debugging across Arize Phoenix and GCP Cloud Trace.I build AI that is observable, testable, and production-hardened.Explore my Digital Twin and Technical Portfolio here: <https://priyaraman-ai-portfolio-76375761518.us-central1.run.app>

Experience

Cyclotron, Inc.

AI Engineering Architect

June 2026 - Present (4 months)

Agentic Solution Architecture design document Review & Governance Framework | Azure AI Foundry, Anthropic Claude, Azure API Management (APIM), Application Insights, Azure Blob Storage, Azure AI Content Safety Guardrails, Prompt Caching, Prompt Governance

Automated Design document scoring system for a Bank Client, evaluating design submissions at SDLC quality gates.

- Prompt Cache Optimization: Implemented Anthropic prompt-caching (ephemeral breakpoints, warm-cache priming, Turn-2 output capping, Caching between Turn1 and Turn2 as history gets resent) across the full design document context — measured cache-hit rates up to 71.8% (64.6% mean across 33 runs), materially cutting per-run token cost.
- Telemetry & Session Cost Visibility: Instrumented Azure Application Insights feeding a live HTML dashboard (API calls, cache performance, domain breakdown, borrowed-evidence tracking), plus a zero-API-call pre-flight cost estimator accurate to ~2% of the actual invoice.
- Two-Pass Orchestration & Model Routing: Architected a two-pass pipeline — Opus evidence extraction (Pass 1) delegating to 7 parallel per-domain sub agent Sonnet scoring calls (Pass 2); implemented dual-layer, Design document-metadata-driven model routing and APIM subscription-key/JWT-based inter-service authentication.

- Anti-Hallucination & Evidence Traceability: Enforced strict JSON-only, evidence-only outputs in Pass 1 and sandboxed each Pass 2 domain to its own evidence plus a deterministic cross-domain backfill; linked every rubric criterion's score back to its supporting source evidence.
- Config-Driven Governance: Eliminated 17+ hardcoded values by splitting the EA-owned scoring rubric from engineering config; established versioned, immutable prompt governance with audit-trail stamping, and evaluated AI/ content-safety guardrail enforcement across four candidate layers.

Self-Employed

Generative AI Architect & Engineer

March 2025 - May 2026 (1 year 3 months)

Building production-grade GenAI systems spanning multi-agent orchestration, LLM fine-tuning, and cloud-native LLMOps.

Distributed Multi-Agent Research Orchestrator | ADK 1.20, LangGraph, LlamaIndex, MCP, GoogleGenAI Embedding, FAISS

- Production-ready multi-agent system structured for Vertex AI Agent Engine
- "Telemetry Gravity" pattern: unified observability across 3 instrumentation libraries
- Custom MCP Server: <2 min podcast generation (80% complexity reduction)
- 91% eval pass rate via Custom Manual Rubric Evaluators
- Advanced RAG: RRF, hybrid search[BM25+Vector Search], hierarchical chunking, FAISS, GoogleGenAIEmbedding

Specialized Drug Interaction LLM | Llama-3.1-8B, PEFT/LoRA, Vertex AI, Cloud Run, GCS

- 331% performance gain via domain-specific fine-tuning with Unsloth/LoRA
- Production 3-tier GCP: Vertex AI Endpoints + Cloud Run + API Gateway
- Full MLOps: LLM-as-a-Judge, GGUF quantization, Docker, deployment automation

Generative Bid Agent | Vertex AI Search, Gemini Pro, Next.js, GCS, Docker

- Multi-modal RAG synthesizing PDF/Excel/PPT for automated RFP responses
- 2-stage retrieval with dynamic filters + Vertex AI Ranking API

Priya's AI Portfolio & Reasoning Engine (The Digital Twin) | Next.js 15, Gemini 2.5 Flash, Cloud Run, Docker

- Agentic Digital Twin & AI Guardrails: Synthesized 18+ years of history into structured JSON memory with strict refusal heuristics to ensure 100% professional integrity.
- Cloud Architecture & FinOps: Deployed to Cloud Run with secure API key injection (Secret Manager) and scale-to-zero cost optimization.
- Explore my digital Twin and Technical Portfolio here: <https://priyaraman-ai-portfolio-76375761518.us-central1.run.app>

GAIA Agent | LangGraph, Mem0, Gemini API

- 100% GAIA benchmark accuracy via multi-step tool-use architecture
- 12 integrated tools with Mem0 persistent memory

KSAP Technologies, Inc.

Generative AI Architect & Engineer

February 2024 - March 2026 (2 years 2 months)

Aurora, IL

Leveraging Google ADK, Agent Engine, GCP, Cloud Run, and LlamaIndex to deliver production-grade enterprise automation.

- Autonomous Ingestion & Cloud Migration: Engineered autonomous agents to automate extraction from complex domain-specific documents and legacy-to-cloud migrations with 100% logic adherence.
- Self-Healing Agents & Unified Observability: Deployed fault-tolerant, self-monitoring QA agents and unified telemetry to ensure enterprise-wide deployment readiness.

CVS Health

Senior Manager, Architecture / Digital Principal Engineer I

October 2021 - October 2023 (2 years 1 month)

Northbrook, IL

- Architected and designed end-to-end digital solutions for enterprise iOS and Android applications, collaborating with backend microservices from conceptualization and estimation to final delivery.
- Led security enhancements by researching and prototyping modern authentication standards. Implemented passwordless solutions (WebAuthn/FIDO2, Apple Passkeys), OAuth 2.0/OIDC with Auth0 (PKCE), and biometric identity verification using FaceTec to prevent account takeovers.
- Drove significant app performance optimization, reducing launch time in low-network conditions from 9s to 300ms.

- Expertly utilized Xcode Instruments & Diagnostics for performance tuning, profiling memory leaks, resolving thread concurrency, debugging memory graphs, and analyzing network behavior.
- Developed modular iOS SDKs and implemented key features like MFA OTP and user session security enhancements.
- Translated business requirements into technical architecture plans, managed domain roadmaps, and presented solutions to stakeholders and upper management.
- Conducted root cause analysis for production issues, performed impact analysis for framework migrations, and resolved critical crashes identified via New Relic.
- Oversaw code quality by leading pull request reviews, enforcing coding guidelines and best practices, and mentoring junior team members within a CI/CD environment.

Capgemini

Solutions Architect

January 2019 - October 2021 (2 years 10 months)

- Architecting solutions for new initiatives for iOS and Android applications collaborating with Orders based micro services in Backend whilst supporting the critical production issues and market defects for 5 regions.
- Conceptualization, design, estimation and overall delivery, Technical architecture & documentation
- Proficient in design and development of App layer code and Inhouse iOS SDKs, Dependencies involving Modularized approach for each product component, Generics , Networking Queues, Closures, Extensions, Protocols&Extensions, Error Handling, Networking.
- Gain an indepth understanding on accessory tools such as Code review process, Unit testing process, Continuous Integration/Continuous Deployment process and security/vulnerability concerns.
- Engage in Pull Request Review process, providing comments on Architecture and Coding guidelines by adhering to best practices in Human Interface Guidelines and language.
- POCs on New features & frameworks and their adoption leading to enhanced features and qualitative user experience.
- Presentation to Product Owners and upper management on offloading Client side Logic to Backend
- DAST – Mobile Security - Revisited and implemented End to end solutioning for preventing insecure session timeout

- Incorporated service based authentication mechanisms using OAuth2.0 standards for authentication and authorization flows within the App
- Analysis and remediation of Veracode security vulnerability reports
- Responsible for Apple sign in implementation, Security enhancements w.r.to user's session, Ordering redesign, Enhancing mobile app for New Products in Menu.
- Proficient in App Performance tuning, Thread Concurrency and race conditions, Leveraged Instruments & Diagnostics
- Experience in Offline Mobile storage – Realm and app upgrade scenarios – Realm migration.
- Troubleshooting and fixing critical showstopper crashes identified from New Relic

Allstate

iOS Developer

July 2018 - December 2018 (6 months)

Greater Chicago Area

About Drivewise module :

Provides personalized assessment of user driving patterns and advocates for safe driving by offering policy discounts.

✓ Dashboard with My Driving assessment by recording Trips, my challenges and Rewards earned for safe driving along with Policy discounts

My Contribution:

- Improved Dashboard performance by more than 75% with Concurrent webservises execution instead of Serial execution.
- Enhanced the Dashboard Architecture from MVC to MVVM using Property Observers.
- Integrated Adobe In App Messaging – To Activate Drivewise Module for Prospective Eligible Customers.
- Integrated Urban Airship – namedUser&ChannelId Integration through Mobile Services and Native iOS SDK, Captures custom Events & Tags, send out push notifications based on segments with Tags, and Automations.

Keevat LLC

iOS Developer

April 2015 - June 2018 (3 years 3 months)

Woodridge IL 60517

** Integrated RESTful services from Yelp, Food And Drug Administration, National Library of Medicine, GoodRx, CMS and Walgreens.

- ** Analyzed Crash reports from dsym, and fixed issues such as the one with WatchDog timers which comes up in TestFlight and App Store builds and also, through Firebase crashAnalytics, analyzed the crashes happening in App.
- ** Integrated User Location Tracking from App2 for the user of App1 by using Core Location APIs and Firebase cloud messaging. So, on click of Track by User in App2, App1 User can be tracked. Used Realm as the backend.
- ** Emergency SOS call can be placed for users in App2 by App1 User and their updated Location Info will also be sent.
- ** Medication Based reminders for all Slots in a single day can be created using EventKit in single creation, app will take care of reminding when next slot is up instead of showing up next day same time slot.
- ** Walgreens Refill can be done online and refill will be ready.
- ** App Widget in Today's Extension for Emergency SOS Contacts

In Firebase,

- ** Integrated Push Notifications in Firebase from App 1 to App 2, notifications are fired from App1 and fired to App2.
- ** Created Dynamic Links - and allowed the link to be passed from App1 to App2 - which demands App2's installation along with Pertinent details about User in App1.
- ** Used Authentication Mechanism in Firebase to create users and maintain their registration.

Wellmark Blue Cross and Blue Shield

Product Analyst Technical

April 2014 - April 2015 (1 year 1 month)

Cognizant Technology Solutions

Associate

September 2006 - December 2012 (6 years 4 months)

Chennai, India

Senior Facets Application Developer/ Tech Lead

- * Expertise in Core Facets Modules - Claims, Members, Provider, Plan & Product, Accumulators, UM&UR and Core Batches

- * Experienced in Major Capital Project Implementations - EPS Implementation, ICD10 implementation, UM&UR Vendor Implementations, Claims Highway, Generic Provider Loads

- * Experienced in Project Management activities such as Project Plan, Estimation Submission, Code Review, Software Configuration Management and Defect Prevention

Education

The University of Texas at Austin

Post Graduate Program, Artificial Intelligence · (July 2024 - April 2025)

Anna University

Bachelor of Engineering (B.E.), Electrical, Electronics and Communications
Engineering · (2002 - 2006)